

Financial Time Series Prediction Using Transformers



Course ID: **CS3103**
Group ID : **25SK18**

Member Names :
Rahul Jilowa (2301AI18)
Chahat Mahajan (2301AI49)
Tripti Jain (2301CS60)

PREDICT BETTER, GROW SMARTER



Roles & Responsibilities

1

TRIPTI JAIN

Led data collection and preprocessing. Assisted with literature review, data visualization and analysis.

2

CHAHAT MAHAJAN

Led literature review, data visualization and analysis. Assisted with data preprocessing and collection.

3

RAHUL JILOWA

Led model implementation. Designed the basic model architecture and trained it to yield the best possible results.



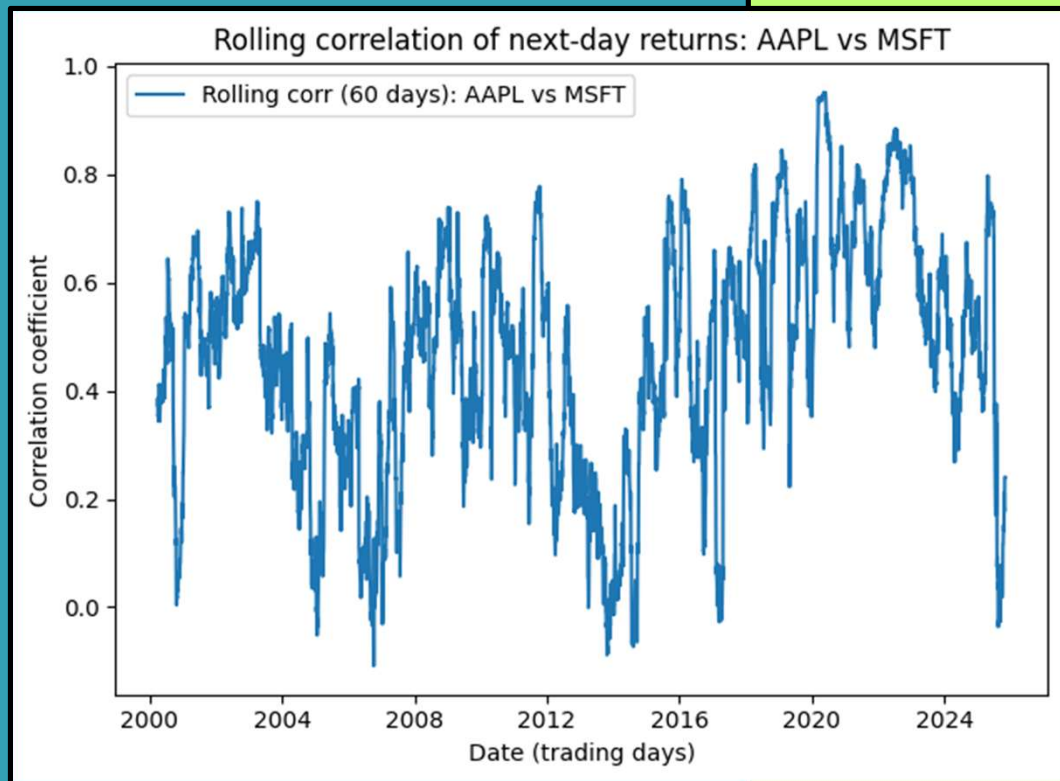
INTRODUCTION

FINANCIAL TIME SERIES PREDICTION USING TRANSFORMERS

Financial time series prediction is difficult due to market noise, volatility, and non-stationarity. Classical models like ARIMA or regression fail to capture temporal dependencies, while RNNs and LSTMs suffer from limited context.

Motivation:

Transformers leverage self-attention mechanisms that enable modelling global temporal dependencies efficiently. Recent works have demonstrated their effectiveness for financial data prediction. Our goal is to evaluate the predictive capability of attention-based models and try to improve the results.



*From our analysis and visualization

OBJECTIVES

1

Evaluate the predictive capability of attention-based models

2

Try to improve the results of previous models by working on shortcomings

3

Analyse and exploit inter-ticker dependencies

4

Demonstrate the use of transformers on S&F500 stocks over 6500 days

LITERATURE REVIEW

Gaps:

1. The study excludes transformer-based or attention-driven architectures, limiting exploration of modern sequence models for nonlinear nonstationary forecasting.
2. The paper lacks empirical comparison with hybrid CNN-LSTM or ensemble models integrating traditional and transformer approaches.
3. The model overlooks nonlinear dependency handling between liquidity indicators, relying solely on linear ridge regularization for ensemble stabilization.
4. The study enhances attention efficiency but does not assess how ridge-based stabilization interacts with transformer attention weights under volatile financial data.

Key Prior Work and Citations:

1. **Chishti, S.M. (2025). Hybrid Deep Learning Architectures for Time Series Forecasting. International Journal of Engineering & Extended Research**
 - Discusses CNN-LSTM hybrids (no transformers). Shows ARIMA underperforms in nonlinear data, while CNN-LSTM adapts to non stationarity
2. **Adapala, N., & Shabbir, A. (2025). Artificial Intelligence in Stock Market Prediction: A Deep Learning Perspective. NHS Journal of Science.**
 - Comprehensive analysis showing that Transformer models handle multi-scale volatility better than ARIMA or LSTM
3. **Yaşar et al. (2025). Reliable Hybrid Forecasting for Financial Liquidity. IEEE Conference.**
4. **Feng & Sinchai (2025). Attention-Revamped Transformer. Arabian Journal of Science and Engineering.**
 - Ridge regression stabilizes ensemble predictions by penalizing correlated base learners.
5. **A. Vaswani et al., "Attention is All You Need," NeurIPS, 2017.**

METHODOLOGY - Dataset

Dataset

We have used **yahoo finance** dataset for extracting real time data. We downloaded the dataset and cleaned it

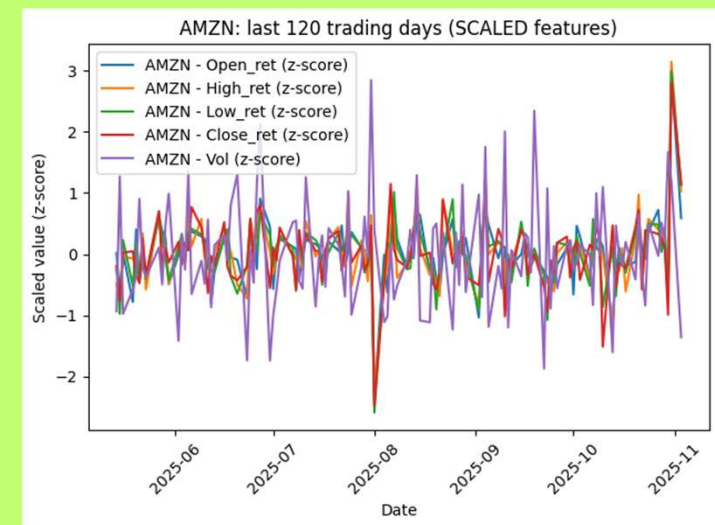
Data Preprocessing

We preprocessed the data to use only open, high, low, close and volume returns. We also reformatted the data to suit the model requirements. We used feature correlation heatmaps etc to deduce the final features to be used.

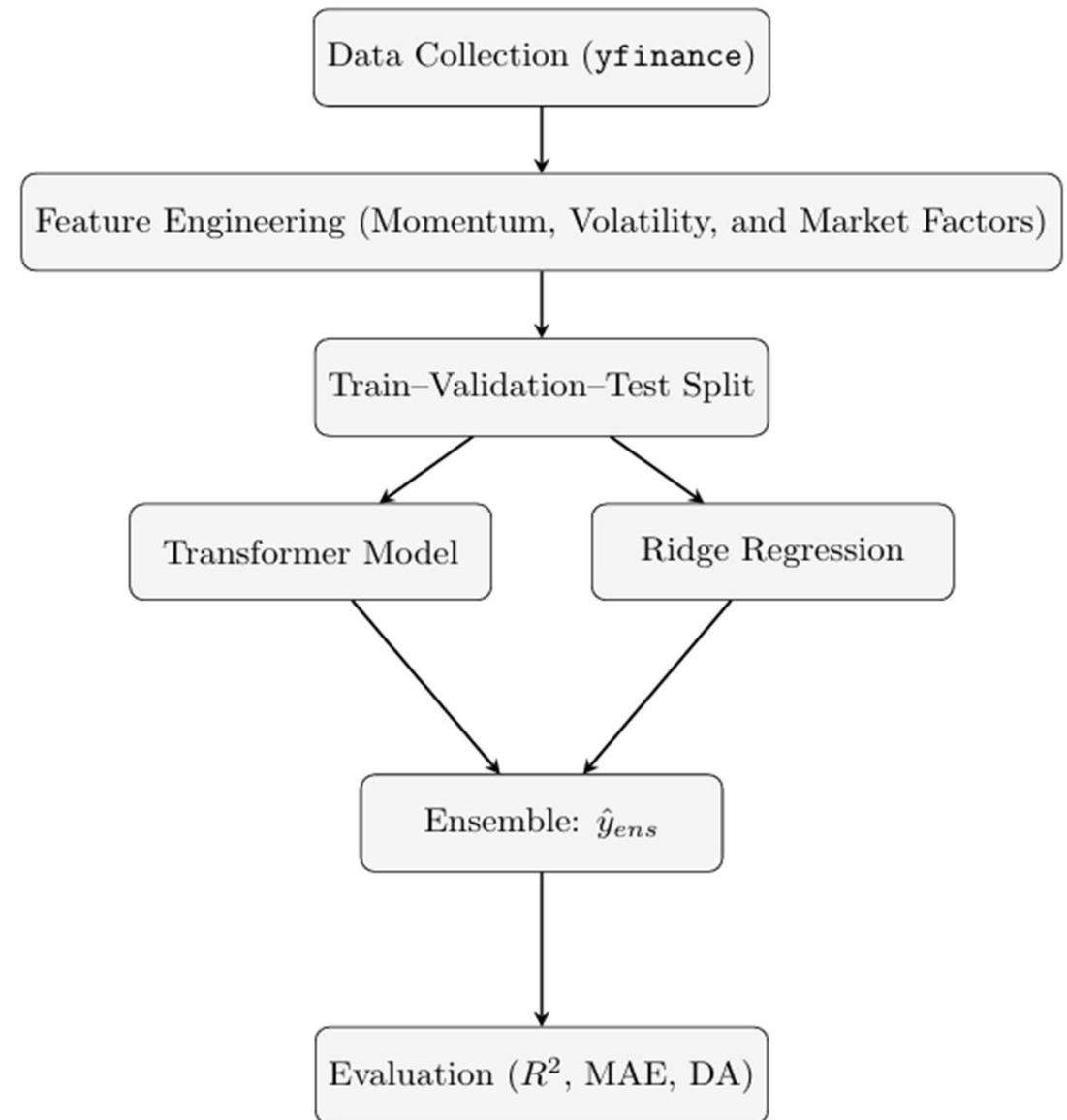
Data Analysis and Visualization

The data was analyzed based on correlation, raw vs scaled data, window sizes, heatmap of cross ticker correlation of next day returns etc. We deduced the interdependencies of various tickers to be useful for prediction, refining the preprocessing part, transformer techniques to use etc.

	Date	AMZN_Open	AMZN_High	AMZN_Low	AMZN_Close	AMZN_Volume	COST_Open	COST_High	COST_Low	COST_Close	...
2	2000-01-03	4.075000	4.478125	3.952344	4.468750	322352000	30.418460	30.418460	29.374587	29.729504	...
3	2000-01-04	4.268750	4.575000	4.087500	4.096875	349748000	29.708631	29.771264	28.038434	28.101067	...
4	2000-01-05	3.525000	3.756250	3.400000	3.487500	769148000	28.205447	28.685629	27.954918	28.581242	...
5	2000-01-06	3.565625	3.634375	3.200000	3.278125	375040000	28.936169	29.771267	28.894414	29.155382	...
6	2000-01-07	3.350000	3.525000	3.309375	3.478125	210108000	29.019663	31.149163	28.998785	31.086531	...



SYSTEM FLOW CHART



METHODOLOGY - System Architecture

Proposed Method

To improve multi-asset financial forecasting, we propose a **hybrid ensemble approach** that combines a **Transformer model** for capturing complex temporal dependencies and inter-stock relationships with **per-ticker Ridge regressors** that learn local linear behaviors. The ensemble balances global sequence learning and individual asset stability, aiming to enhance both **directional accuracy** and **robustness** of predictions.

$$\hat{y}_{ens} = \alpha \hat{y}_{trans} + (1 - \alpha) \hat{y}_{ridge}.$$

Model Architecture

1. Transformer Forecaster:

A sequence-to-sequence architecture with

- **Input:** 30-day rolling window of engineered features per ticker.
- **Embedding Dimension (d_m):** 256, with 8 attention heads and 6 encoder layers.
- **Feedforward Layers:** dimension 512, dropout 0.1, trained for 80 epochs using Adam ($\text{lr} = 3 \times 10^{-4}$).
- Captures **temporal dependencies** and **cross-ticker relationships** in financial data.

2. Per-Ticker Ridge Regressors:

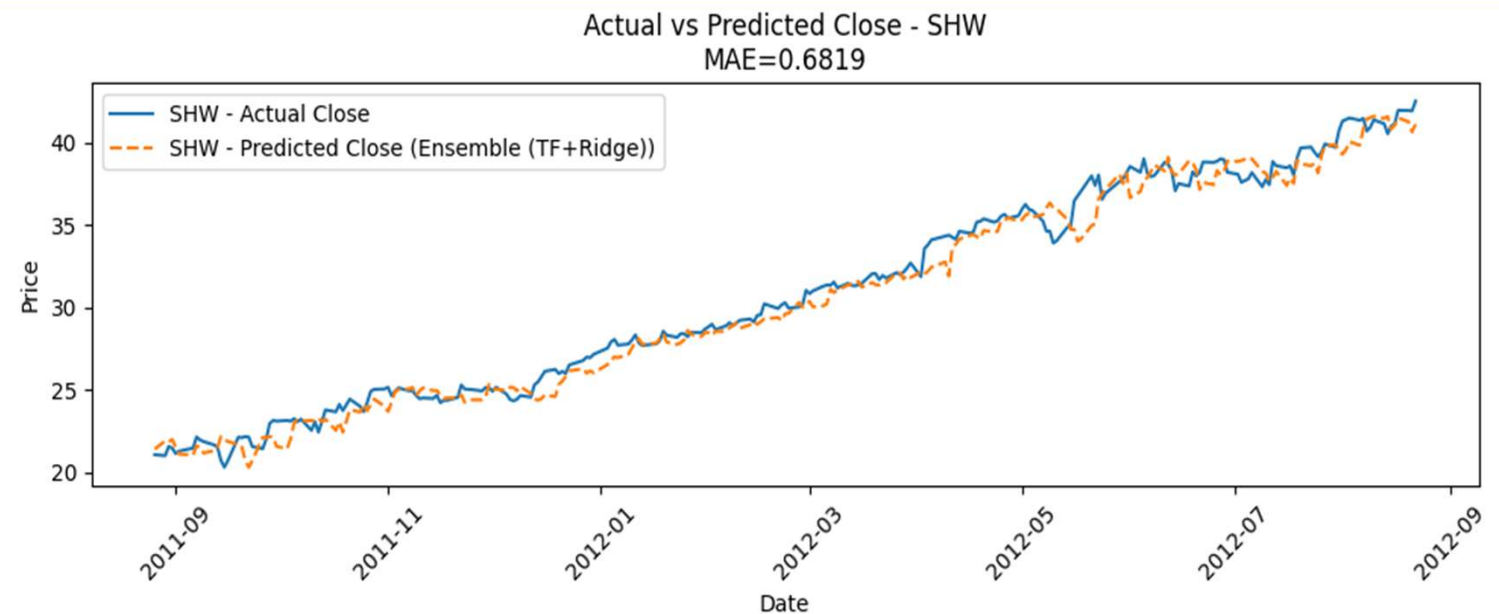
- Linear models trained **independently for each ticker** on the same features.
- Capture **local, short-term linear trends** and reduce overfitting on individual assets.

RESULTS & DISCUSSIONS

- Ensemble improved **DA** by **~1.1%** over transformer-only model.
- **Return R^2** values were near zero or slightly negative — typical for financial returns (high noise).
- **Price-level $R^2 > 0.9$** , showing that model tracks overall trends well.

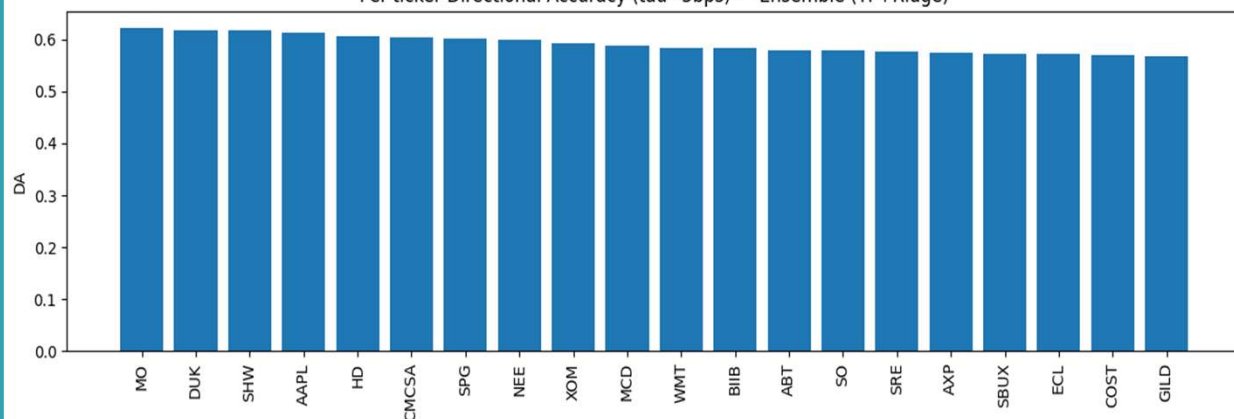
Discussion Points

- **Transformer strength:** Captured long-range dependencies better than traditional models.
- **Ridge contribution:** Stabilized noisy predictions and exploits inherent linearity.
- **Directional Accuracy meaning:** Slightly above random (50%) → statistically small but *non-random signal* in market data.
- **Interpretation:** Market returns are inherently noisy and near-unpredictable, so even small improvements are meaningful.



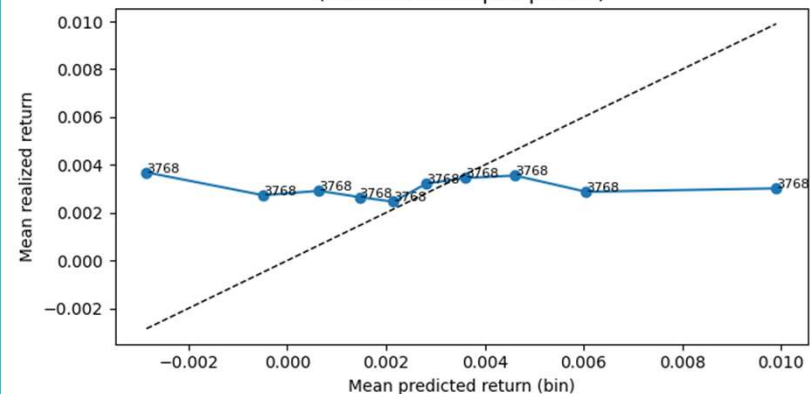
This plot compares actual closing prices with the model's predicted values for SHW. The closer the dashed line (predictions) follows the solid line (actuals), the more accurate the model's forecasts are for SHW.

Per-ticker Directional Accuracy (tau=5bps) — Ensemble (TF+Ridge)



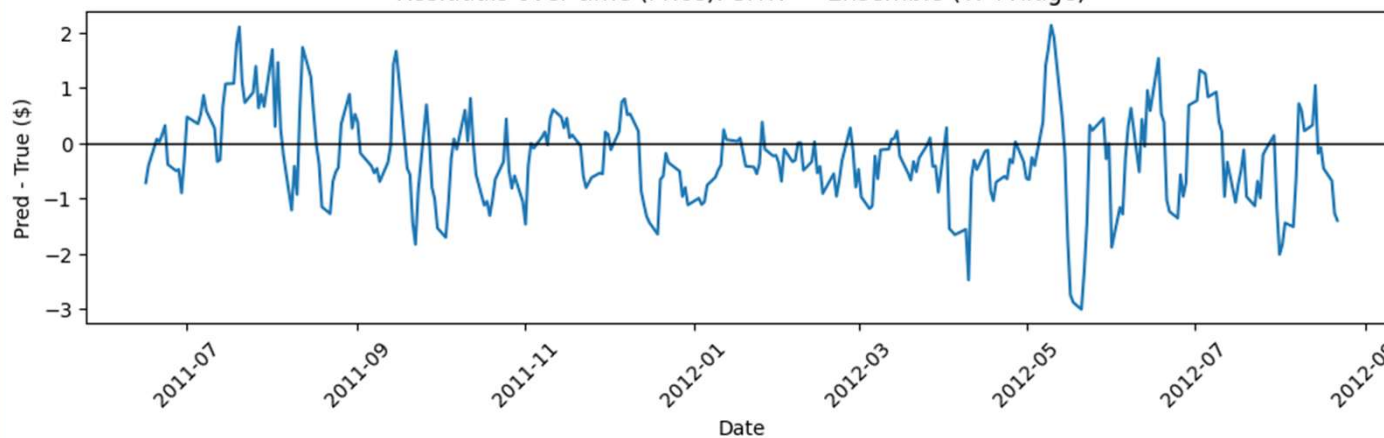
This bar chart shows DA values for each ticker. Higher R^2 (or lower MAE) indicates better model performance for that specific stock.

Return calibration (deciles) — Ensemble (TF+Ridge)
(numbers = samples per bin)



This plot tests if predicted return magnitudes align with actual outcomes across deciles. A close match along the diagonal indicates strong calibration.

Residuals over time (Price): SHW — Ensemble (TF+Ridge)



This time series of residuals for SHW helps detect drift or bias. If errors consistently stay above or below zero, the model may be systematically off for SHW.

KEY OBSERVATIONS

- 1 A larger dataset, with more frequent data would have helped train the transformer better
- 2 Ensemble outperformed the Transformer, suggesting lower noise robustness and underlying linear trends in the data.
- 3 Stocks have slight interdependencies which can be exploited to yield better results.
- 4 Market uncertainties cannot be predicted just using OLHCV features. We need to take into account market sentiment and daily affairs
- 5 The data is not very stationary which makes it hard for a small model to predict effectively

CONCLUSION

Limitations :

1. Dataset was small
2. Limited GPU access
3. Lack of real time and stationary data
4. Sensitive to label scaling
5. Assumes residuals are IID and Gaussian

Future Work:

- Adding an extra layer to study the trends in our time series. ARIMA or LSTM-GRU can be pretty good approaches here
- We can also add a CNN model to capture trading signals which can help with better predictions especially for commodities.
- We can analyse regular market sentiments, government policies, global situation etc by combining it with a natural language processing model.

REFERENCES

1

A. Vaswani et al., "Attention is All You Need," NeurIPS, 2017.

2

P. Brugi`ere, "Transformer for Time Series: An Application to the S&P500," 2023.

3

G. Turinici and J.P. Morgan, "Financial Time Series Forecasting Using CNN and Trans former," 2023

4

H. Wu et al., "Informer: Beyond Efficient Transformer for Long Sequence Time-Series Fore casting," AAAI, 2021

5

S.M. Chishti, "Hybrid Deep Learning Architectures for Time Series Forecasting," 2025

6

N. Adapala and A. Shabbir, "Artificial Intelligence in Stock Market Prediction: A Deep Learning Perspective" 2025

7

H. Ya,sar et al., "Reliable Hybrid Forecasting for Financial Liquidity," IEEE Conference, 2025

8

J. Feng and S. Sinchai, "Attention-Revamped Transformer," Arabian Journal of Science and Engineering, 2025

THANK YOU

Code Link: <https://colab.research.google.com/drive/11O7uJuMAVLLkjo5xlbFkF8n08-cWLbXs?usp=sharing>

GitHub Link: <https://github.com/tripsycodes/Financial-Time-Series-Prediction-Using-Transformers>